

HAAPAVEST MUSTIKKAMAKI, Finland

WMO: 027970

Lat: 64.142N

Lon: 25.423E

Elev: 112

StdP: 99.99

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.5	(c) -23.7	(d) -29.2	(e) 0.3	(f) -26.4	(g) -26.2	(h) 0.4	(i) -23.7	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 27.1	(d) 18.6	(e) 24.9	(f) 17.2	(g) 22.9	(h) 16.2	(i) 19.6	(j) 24.6	(k) 18.4	(l) 23.1	(m) 17.3	(n) 21.5	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.1	21.7	16.6	12.0	20.3	15.4	11.1	19.3	56.3	24.9	52.6	23.0	49.0	21.7	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -27.7	(f) 29.0	(g) 4.7	(h) 2.9	(i) -31.0	(j) 31.1	(k) -33.8	(l) 32.8	(m) -36.4	(n) 34.4	(o) -39.8	(p) 36.5		
(a) N/A	(b) N/A	(c) N/A	(e) -27.8	(f) 21.0	(g) 4.6	(h) 2.1	(i) -31.2	(j) 22.5	(k) -33.9	(l) 23.8	(m) -36.5	(n) 24.9	(o) -39.8	(p) 26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-9.9	-7.8	-4.1	2.0	9.0	13.0	16.4	14.2	9.5	2.7	-1.2	-5.4
	DBStd	9.94	7.22	7.11	4.98	3.65	4.66	3.94	3.49	3.25	3.42	3.70	4.49	6.28
	HDD10.0	2974	617	498	436	240	74	14	1	4	50	226	337	477
	HDD18.3	5532	875	732	694	489	290	167	81	133	266	483	587	735
	CDD10.0	515	0	0	0	1	43	103	199	134	34	1	0	0
	CDD18.3	32	0	0	0	0	1	6	20	5	0	0	0	0
	CDH23.3	364	0	0	0	0	33	71	209	51	0	0	0	0
	CDH26.7	70	0	0	0	0	2	12	49	7	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	556	37	31	26	26	45	59	76	67	54	48	46	43
	PrecMax	699	70	57	49	50	79	119	144	146	114	90	87	91
	PrecMin	410	12	6	5	10	9	19	17	16	18	5	6	14
	PrecStd	76	15	13	12	12	19	28	28	33	26	24	19	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	4.8	7.7	18.1	26.2	28.6	30.2	27.8	20.7	12.6	8.9	4.1
		MCWB	2.4	3.0	3.9	9.7	15.0	17.9	19.5	20.4	15.7	11.0	8.0	3.6
	2%	DB	1.2	3.2	5.5	13.3	24.0	25.5	28.1	24.5	18.1	10.6	6.7	2.7
		MCWB	0.7	1.9	2.6	7.3	14.2	17.6	19.2	17.9	14.1	9.4	6.1	2.2
	5%	DB	0.4	1.6	3.5	10.6	21.1	22.7	25.6	22.3	16.2	9.3	5.3	1.7
		MCWB	0.0	1.0	1.5	6.1	13.3	15.3	18.1	16.6	13.1	8.3	4.9	1.4
	10%	DB	-0.6	0.5	2.3	8.3	17.7	20.6	23.5	20.2	14.9	8.0	4.1	0.9
		MCWB	-0.9	0.1	0.8	4.6	11.2	14.6	17.2	15.7	12.6	7.1	3.7	0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	3.3	4.6	10.5	16.7	19.6	21.9	20.9	16.3	11.7	8.1	3.8
		MCDB	2.6	4.5	7.3	15.0	23.2	25.0	27.1	25.9	19.6	12.4	9.0	3.9
	2%	WB	0.9	2.1	3.0	8.2	15.2	18.3	20.1	18.9	14.8	9.7	6.0	2.3
		MCDB	1.1	2.9	4.5	12.7	21.9	23.6	25.2	23.4	17.1	10.2	6.5	2.5
	5%	WB	0.1	1.1	1.9	6.3	13.6	16.7	19.0	17.5	13.8	8.5	5.0	1.3
		MCDB	0.4	1.6	3.3	9.7	20.3	21.4	24.0	20.8	15.6	9.2	5.3	1.6
	10%	WB	-0.9	0.2	1.0	4.9	11.8	15.1	17.9	16.4	12.7	7.1	3.8	0.7
		MCDB	-0.6	0.5	1.9	7.8	17.2	19.4	22.4	19.4	14.4	7.9	4.0	0.9
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.2	8.3	9.0	10.8	9.9	9.9	9.4	7.7	5.3	3.7	4.5
		MCDBR	4.2	4.8	8.3	13.6	16.1	14.5	14.0	13.0	10.5	6.8	4.3	3.9
		MCWBR	4.2	4.1	6.2	8.4	8.2	7.4	6.7	7.2	6.9	5.7	4.2	3.7
	5% WB	MCDBR	4.3	4.5	6.9	12.3	14.7	11.7	11.9	10.8	9.5	6.3	4.4	3.6
		MCWBR	4.3	4.0	5.5	8.0	8.0	6.5	6.3	6.5	6.6	5.6	4.3	3.5
Clear-Sky Solar Irradiance	taub	0.214	0.279	0.308	0.348	0.344	0.347	0.368	0.350	0.327	0.306	0.355	0.356	
	taud	1.976	2.127	2.187	2.304	2.447	2.462	2.421	2.461	2.515	2.379	2.622	2.643	
	Ebn at Noon	348	622	771	826	868	872	838	821	768	607	322	172	
	Edh at Noon	27	62	89	101	96	97	98	86	67	49	24	15	
All-Sky Solar Radiation	RadAvg	0.14	0.69	2.16	3.74	4.90	5.42	5.03	3.79	2.14	0.80	0.18	0.05	
	RadStd	0.01	0.13	0.21	0.34	0.52	0.46	0.49	0.38	0.22	0.14	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page